

EE 527P: Practicum on Analysis and Design of Power Electronic Converters

Course Code : EE 527P

Course Name: Practicum on Analysis and Design of Power Electronic Converters

L-T-P-C: 0-0-3-2

Prerequisites: EE309 and EE309P- Power Electronics or Equivalent

Intended for: BTech final year and M.Tech in Power Electronics and Drives (PED)

Distribution: Core for M.Tech. in PED, Elective for final year Electrical Engineering and other PG

Approval: 13th Senate

Course Contents

This is a laboratory course with 3-hour sessions per week. Following is the list of course modules and experiments.

- **Part I** Part-1 will involve experiments related to analysis, design, control and applications of power electronic converters
 - Basic and few advanced AC-DC, DC-AC, DC-DC and AC-AC converters: Analysis and design (11 hours)
 - Closed loop control of power electronics converter (10 hours)
- Demonstration of few power electronics converter applications such as renewable grid interactive converter, power factor correction circuits etc. (6 hours)
- **Part-II** (15 hours) Part-II will involve a design project related to analysis, design and control of power electronic converters

Textbook:

1. Lab experimental manuals will be provided for Part-1.

4. References:

1. Mohan N., Undeland T.M. and Robbins W.P., **Power Electronics – Converters, Applications and Design**, 3rd Edition, Wiley India, 2008
2. L. Umammd, **Power Electronics: Essentials & Applications**, Wiley India, 2009.
3. R.W. Erickson, D. Maksimovic, **Fundamentals of Power Electronics**, 2nd Edition, Kluwer Academic Publishers.
4. Bin Wu, **High Power Converters and AC Drives**, IEEE Press, John Wiley & Sons, 2006.
5. Rashid M., **Power Electronics - Circuits, Devices and Applications**, 3rd Edition, Pearson Education.
6. A. T.Pressman, **Switch Mode Power Supply Design**, McGraw-Hill, 1999.
7. R.S. Ramshaw, **Power Electronics Semiconductor Switches**, Chapman & Hall, 1993.
8. D. Grahame Holmes, Thomas A. Lipo, **Pulse width modulation for power converters: principles and practice**, John Wiley & Sons, 2003.